

MySQL Interview Questions & Answers

Palle Technologies
Updated on 22-oct-2022

all queries are based on below tables

table 1 - department table

primary key

unique key

dept_id (PK)	dept_name (UK)	dept_location
1	HR	BTM
2	Development	HSR
3	Training	HSR
4	Sales	Bommanahalli
5	Marketing	Bommanahalli
6	MD	HSR
7	Recruitment	HSR

All queries are based on below tables

Primary key

table 2 - **employee** table

foreign key

emp_id (PK)	emp_name	city	emp_salary	manager_id	d_id (FK)
1	Ramesh	mumbai	22000	10	1
2	Suresh	bangalore	25000	11	2
3	Mahesh	chennai	21000	8	3
4	Lokesh	chennai	35000	5	4
5	Veena	delhi	32000	9	4
6	Sravanthi	bangalore	18000	11	2
7	Divya	mysore	15000	9	5
8	Jona	chennai	30000	9	3
9	Steve	chennai	40000	Null	6
10	Nithesh	delhi	35000	9	1
11	Mohan	kolkata	36000	9	2
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coding or query based questions

- dear students mock interviewers can ask any questions based on the concepts being covered as part of the course while taking mock
- we request all students to prepare the all concepts related queries or code properly
- most frequently asked queries or concepts are listed in the next slide. please refer and prepare accordingly.

- sql query for **creating department table** with appropriate data types and primary key and unique constraints
- sql query for **creating employee table** with appropriate data types and primary key and foreign key constraints

- question based on **update query** on the employee table
- question based on **delete query** on the employee table
- question based on **insert query** on the employee table
- questions based on select statement with **where clause** on employee table
- questions based on select statement with **like clause** on employee table
- questions based on **group by** clause on employee table

- questions based on **having clause** on employee table
- questions based on **inner joins** on employee and department table
- questions based on **left outer joins** on department and employee table
- questions based on **self join** on employee table
- questions based on **sub queries**
- questions based on **co-related sub queries** and **nth highest nth lowest** values

1. what is primary key?

- primary key gives uniqueness to the tables rows
- only one primary key is allowed per table
- primary key will not allow null values and duplicate values

2. What is the difference between primary key and unique key?

primary key	unique key
only one primary key is allowed per table.	any number of unique keys are allowed per table
primary key will not allow duplicates values	unique constraint will not allow duplicate values
primary key will not allow null values	unique constraint will allow any number of null values

3. what is foreign key?

- using foreign key constraint we can link 2 or more tables.
- using foreign key we can achieve referential integrity

Syntax

```
Create table table_name  
(  
Col1 datatype ,  
Col2 datatype,  
foreign key(col1) references table_name(column_name)  
)
```

5. what is the diff b/w char and varchar?

char is a fixed size data type & varchar is varying size data type. wastage of memory will be more in char type.

6. what is normalization? and what are the benefits?

normalization is used for eliminating the data redundancy. normalization will eliminate the insert, update anomalies.

7. is it possible to store null in a foreign key column?

Yes.

8. Is null allowed in primary key column?

no.

9. how many primary keys we can give for a table?

only one.

10. how many unique keys allowed for a table?

any number.

11. How many foreign keys allowed per table?

we can create any number of foreign keys per table.

12. what is the diff between drop and delete?

drop	delete
drop table will delete the table (along with data and structure)	a delete statement will delete only data, but table will be retained.
drop is a DDL statement	delete is a DML statement
Drop is not a logged operation and hence not reversible	Delete is a logged operation and hence it is reversible
eg: drop table employee;	eg: delete from employee where emp_id = 1;

13. what is the diff between delete and truncate?

delete	truncate
1. a delete statement will delete only data, but table will be retained.	1. a truncate statement will delete only data, but table will be retained. (just like delete)
2. delete is a DML statement	2. truncate is a DDL statement
3. we can give where clause in delete. Delete operation is slower.	3. we can't give where clauses in truncate. Truncate is faster than delete.
4. we can roll back the delete operation	4. We can't roll back the truncate operation.

14. what is the purpose of joins in sql?

- joins are useful for getting data from 2 or more tables into a result set.

15. what are the different types of joins supported in sql server?

1.inner join

2.outer join

1)Left Outer Join

2)Right Outer Join

3.Self Join

4.Cross Join

16. what is inner join?

Inner join is used for getting matched data from 2 or more tables based on the condition.

17. what is LOJ?

In left outer join data from left table is completely included into the result-set but only matched data from right side table is included. In place of right table data null's are included where there is no match.

18. What is ROJ?

In right outer join data from right table is completely included into the result-set but only matched data from left side table is included. In place of left table data null's are included where there is no match.

19. What is self join?

- 1.Joining a table to itself is called as self join.
- 2.we must use inner join keyword for self join.

20. what is cross join?

cross join will join each row present in a table with all rows present in other table. in simple words it will produce Cartesian product of all rows present in both the tables.

21. what is the purpose of having clause?

using having clause we can filter the records which are produced by group by clause.

22. what is the purpose of order by clause?

Ans : order by clause specifies that a SQL SELECT statement returns rows with sorted order, by the values of given column.

23. what is the purpose of group by clause?

Ans : using group by clause we can group data based on one or more column.

28. write query for finding employee details with first highest salary?

Ans : student should write query

29. write query for finding employee details with second highest salary?

Ans : student should write query

Link for tables

- Student can use the following referenced tables (usually mock interviews we ask questions based on the given tables)
- <https://palle-sql-tables.s3.ap-south-1.amazonaws.com/index.html>